

Discriminant analysis

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Lecture based on **ISLR book** and its accompanying slides



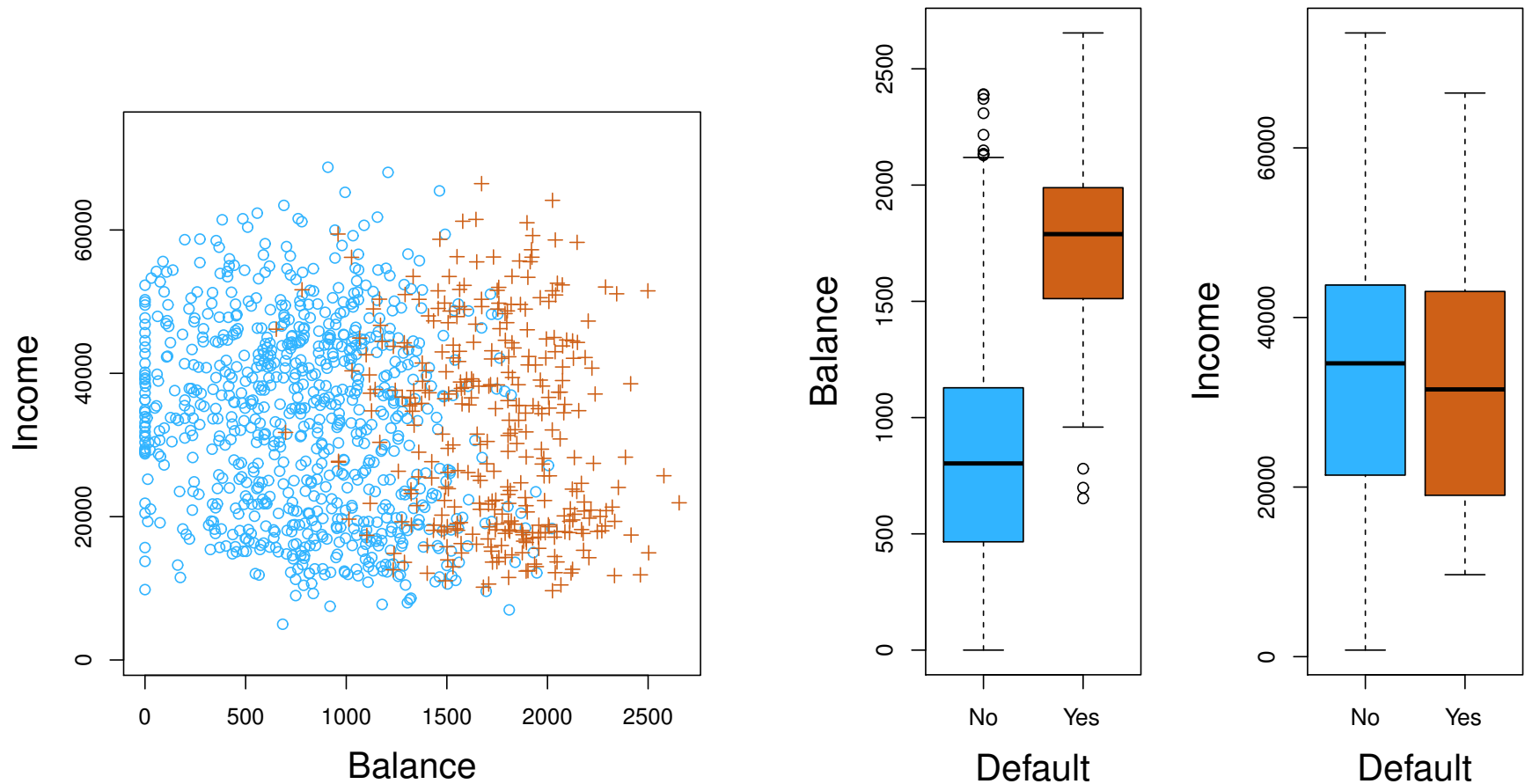
<http://cw.felk.cvut.cz/wiki/courses/b4m36san/start>

Introduction

- Study multivariate relationships with categorical dependent variable
 - independent variables are continuous,
 - but can be categorical too,
 - **nominal** dependent variables take values in an unordered set $\mathcal{C}$
 - * eye color $\in \{\text{brown, blue, green}\}$, email $\in \{\text{spam, ham}\}$,
- the main goals are to
 - **classify** into the target categories
 - * given a feature vector $\mathbf{X} \in \mathcal{X}$ and a nominal response Y taking values in $\mathcal{C}$, the goal is to build a function $f : \mathcal{X} \rightarrow \mathcal{C}$,
 - * often, the mapping is probabilistic $f_p : \mathcal{X} \times \mathcal{C} \rightarrow [0, 1]$,
 - **understand** the role of the individual independent variables
 - * assess the strength of their relationships with the target variable.

Example: Credit Card Default

- Simulated dataset, an individual may default on his credit card payment.



Can we use linear regression?

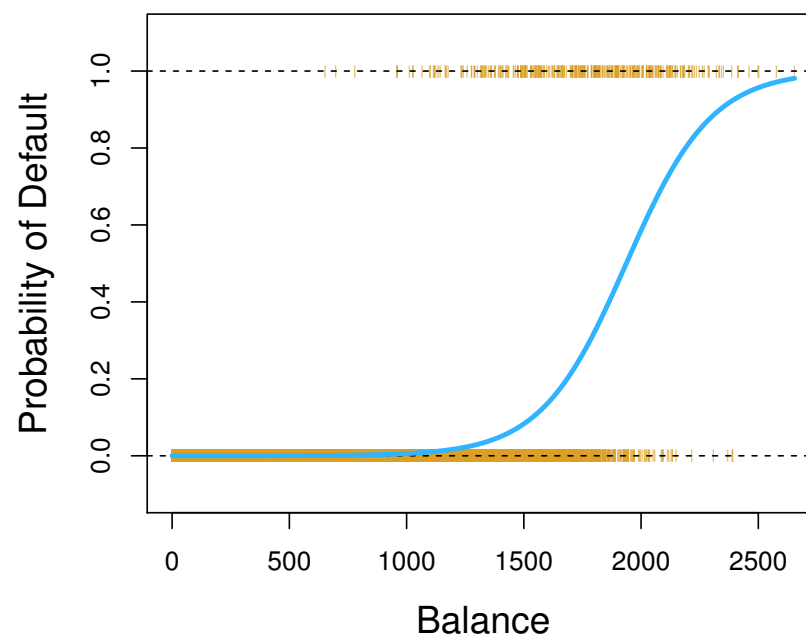
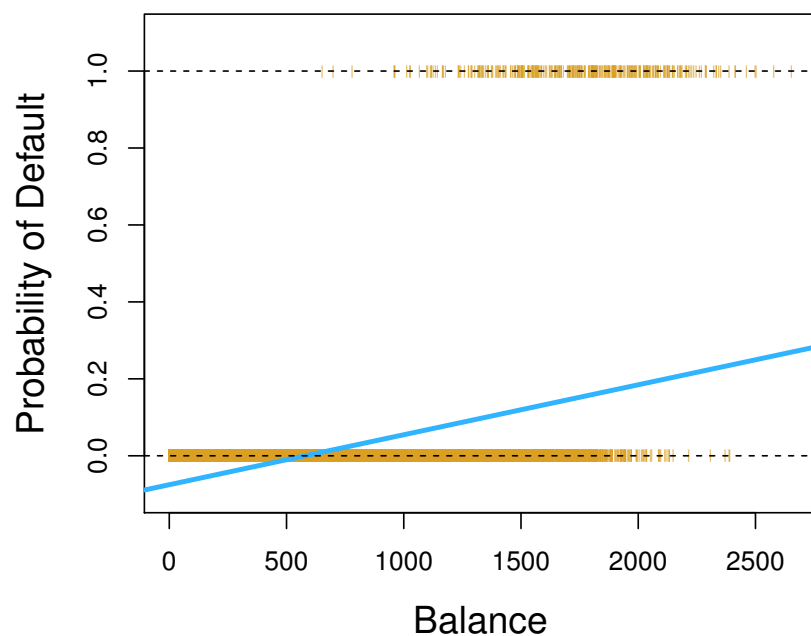
- the target variable Y expressing default can be coded

$$Y = \begin{cases} 0 & \text{if No} \\ 1 & \text{if Yes} \end{cases}$$

- perform a linear regression of Y on X and classify as Yes if $\hat{Y} > 0.5$
 - in this case of a binary outcome, linear regression does a good job as a classifier, and is equivalent to linear discriminant analysis (discussed later),
 - since in the population $E(Y|X = x) = Pr(Y = 1|X = x)$, we might think that regression is perfect for this task,
 - however, linear regression might in general
 - * produce probabilities less than zero or bigger than one,
 - * be sensitive to outliers,
 - * “mask out” some classes in problems with multinomial targets,
- **logistic regression** is more appropriate.

Linear versus logistic regression

- Consider a simple linear model $Y = \beta_0 + \beta_1 \text{Balance}$ (left),
- introduce a non-linear **logit** transformation (right).



The orange marks indicate the response Y , either 0 or 1. Linear regression does not estimate $\Pr(Y = 1|X)$ well. Logistic regression seems well suited to the task.

Logistic regression

- Let's write $p(\mathbf{X}) = \Pr(Y = 1|\mathbf{X})$ for short,
- logistic regression uses the form

$$p(\mathbf{X}) = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)}}$$

- no matter what values β_i or X_i take, $p(\mathbf{X})$ will have values between 0 and 1,
- a bit of rearrangement gives

$$\log\left(\frac{p(\mathbf{X})}{1 - p(\mathbf{X})}\right) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

- this monotone transformation is called the **log odds** or **logit** transf. of $p(\mathbf{X})$,
- we use maximum likelihood to estimate the parameters β_i

$$\ell(\beta_0, \dots, \beta_p) = \prod_{\forall i y_i=1} p(\mathbf{x}_i) \prod_{\forall i y_i=0} (1 - p(\mathbf{x}_i))$$

Logistic regression – motivation

- In linear regression
 - the outcome thresholds the distance to the decision boundary
 - the distance can easily be computed,
- transform this distance to probability $p(X)$ with the following requirements
 - the objects lying on the boundary have $p(X) = 0.5$,
 - distant objects have $p(X) \rightarrow 0$ (in one direction) or $p(X) \rightarrow 1$ (in the other direction),
 - the transformation is most sensitive around the decision boundary,
- transformation steps
 - start with the linear model, its limitations are known,
 - distance has no ceiling $\rightarrow$ turn probability into odds to remove the range restrictions,
 - however, we need to consider direction from the decision boundary too,
 - apply log transform to remove the floor restriction,

Logistic regression – making predictions

- In R `glm` function can be applied to learn logistic models
 - **generalized linear models** allow the linear model to be related to the response variable via a link function, and allow for responses whose error distribution is different from a normal distribution,
- fit the *default* model for *balance*

	Coefficient	Std. error	Z-statistic	p-value
Intercept	-10.6513	0.3612	-29.5	< 0.0001
balance	0.0055	0.0002	24.9	< 0.0001

- fit another *default* model for *student*

	Coefficient	Std. error	Z-statistic	p-value
Intercept	-3.5041	0.0707	-49.55	< 0.0001
student[Yes]	0.4049	0.1150	3.52	0.0004

Logistic regression – making predictions

- Coefficient interpretation in simple models?

- simpler for a binary predictor such as *student* ($Pr(default = y | student = y) = p(s^+)$),
- compare log-odds for the student and non-student groups,

$$\frac{p(s^+)}{1 - p(s^+)} = e^{\hat{\beta}_0 + \hat{\beta}_1} \ \& \ \frac{p(s^-)}{1 - p(s^-)} = e^{\hat{\beta}_0} \rightarrow \frac{\frac{p(s^+)}{1 - p(s^+)}}{\frac{p(s^-)}{1 - p(s^-)}} = e^{\hat{\beta}_1}$$

- $e^{\hat{\beta}_1} = e^{0.4049} = 1.5$ gives the **odds ratio** between the groups,

- where is the decision boundary and what is its shape?

Logistic regression – making predictions

■ Coefficient interpretation in simple models?

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- $e^{\hat{\beta}_1} = e^{0.4049} = 1.5$ gives the **odds ratio** between the groups,

■ where is the decision boundary and what is its shape?

- more clear for a continuous predictor ($Pr(default = y | balance) = p(b)$)

$$p(b) = 0.5 \rightarrow \frac{p(b)}{1 - p(b)} = 1 \rightarrow \log\left(\frac{p(b)}{1 - p(b)}\right) = \beta_0 + \beta_1 b = 0 \rightarrow b = -\frac{\beta_0}{\beta_1} = \$1937$$

Logistic regression – making predictions

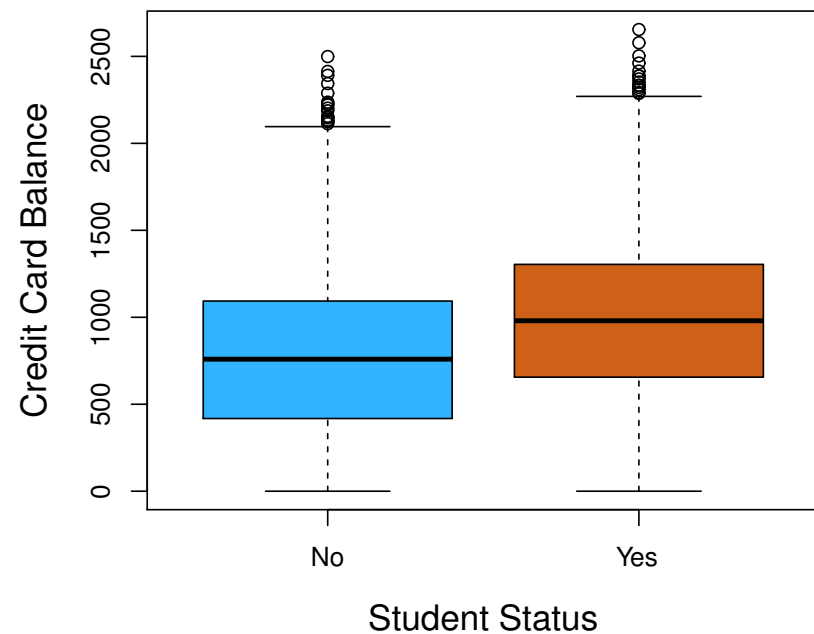
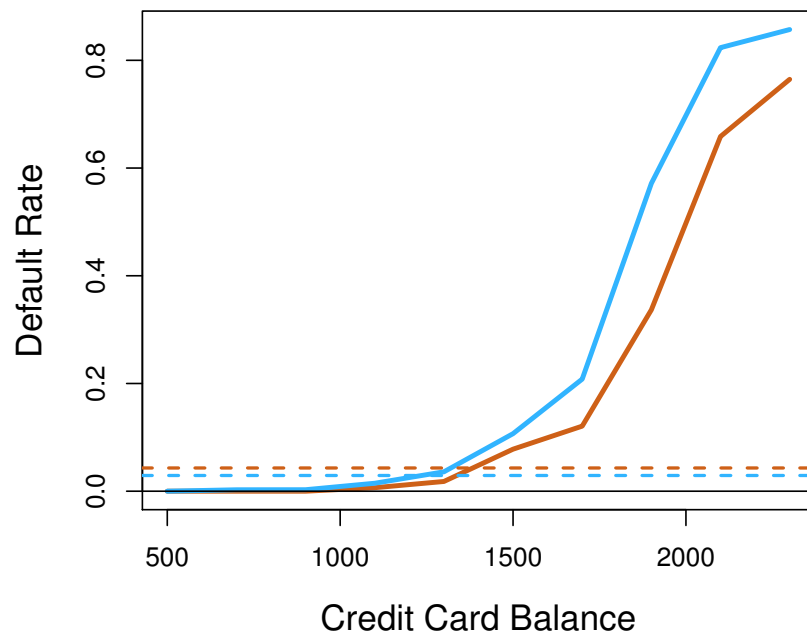
- Now fit the *default* model with several predictors

	Coefficient	Std. error	Z-statistic	p-value
Intercept	-10.8690	0.4923	-22.08	< 0.0001
balance	0.0057	0.0002	24.74	< 0.0001
income	0.0030	0.0082	0.37	0.7115
student[Yes]	-0.6468	0.2362	-2.74	0.0062

- why is coefficient for student negative, while it was positive before?

Confounding

- Students tend to have higher balances than non-students (right)
 - so their marginal default rate is higher than for non-students,
- but for each level of balance, students default less than non-students (left),
- multiple logistic regression can tease this out.



Logistic regression with more than two classes

- So far, logistic regression with two classes only,
- it is easily generalized to more than two classes
 - in symmetric form, there is a linear function for each class

$$Pr(Y = k|\mathbf{X}) = \frac{e^{\beta_{0k} + \beta_{1k}X_1 + \dots + \beta_{pk}X_p}}{\sum_{j=1}^K e^{\beta_{0j} + \beta_{1j}X_1 + \dots + \beta_{pj}X_p}}$$

- this option is used e.g., in the R package *glmnet*,
- in asymmetric form, one of the outcomes is selected as a pivot,
- $K-1$ models are trained

$$\forall i = 1 \dots K - 1 \quad \frac{Pr(Y = i|\mathbf{X})}{Pr(Y = K|\mathbf{X})} = e^{\beta_{0i} + \beta_{1i}X_1 + \dots + \beta_{pi}X_p}$$

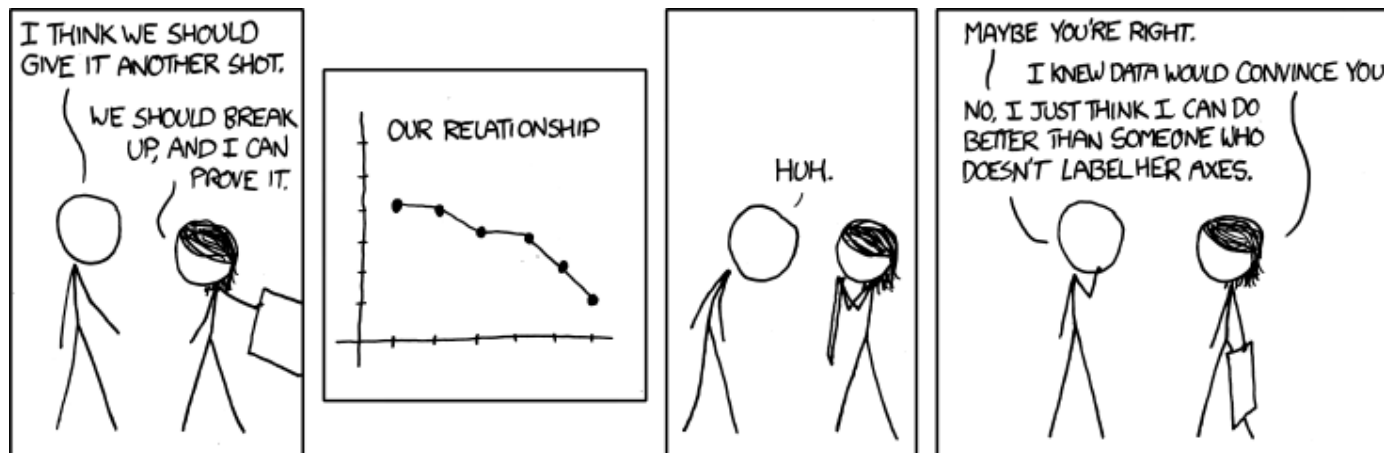
- it can easily be shown that

$$Pr(Y = i|\mathbf{X}) = \frac{e^{\beta_i \cdot \mathbf{X}}}{1 + \sum_{j=1}^{K-1} e^{\beta_j \cdot \mathbf{X}}} \quad Pr(Y = K|\mathbf{X}) = \frac{1}{1 + \sum_{j=1}^{K-1} e^{\beta_j \cdot \mathbf{X}}}$$

- multiclass logistic regression is also referred to as **multinomial regression**.



Scientific comics ...



Taken from <https://xkcd.com>

Discriminant analysis

- The distribution of $\mathbf{X}$ in each of the classes modeled separately,
- Bayes theorem flips things around and helps to obtain $Pr(Y|\mathbf{X})$

$$Pr(Y = k|\mathbf{X} = \mathbf{x}) = \frac{Pr(\mathbf{X} = \mathbf{x}|Y = k)Pr(Y = k)}{Pr(\mathbf{X} = \mathbf{x})}$$

- this approach is quite general,
- when we use normal (Gaussian) distributions for each class
 - this option leads to **linear or quadratic discriminant analysis**

$$Pr(Y = k|\mathbf{X} = \mathbf{x}) = \frac{\pi_k f_k(\mathbf{x})}{\sum_{j=1}^K \pi_j f_j(\mathbf{x})}$$

- where $f_k(\mathbf{x}) = Pr(\mathbf{X} = \mathbf{x}|Y = k)$ is the density for $\mathbf{X}$ in class k ,
- where $\pi_k = Pr(Y = k)$ is the marginal or prior probability for class k .

Linear discriminant analysis for $p=1$

- Plug the Gaussian density model into Bayes formula
($p_k(x) = Pr(Y = k|X = x)$)

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma}\right)^2}}{\sum_{j=1}^K \pi_j \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_j}{\sigma}\right)^2}}$$

- note, that we assume $\forall k \ \sigma_k = \sigma$ here,
- happily, there are simplifications and cancellations,
- maximize the **discriminant score** instead

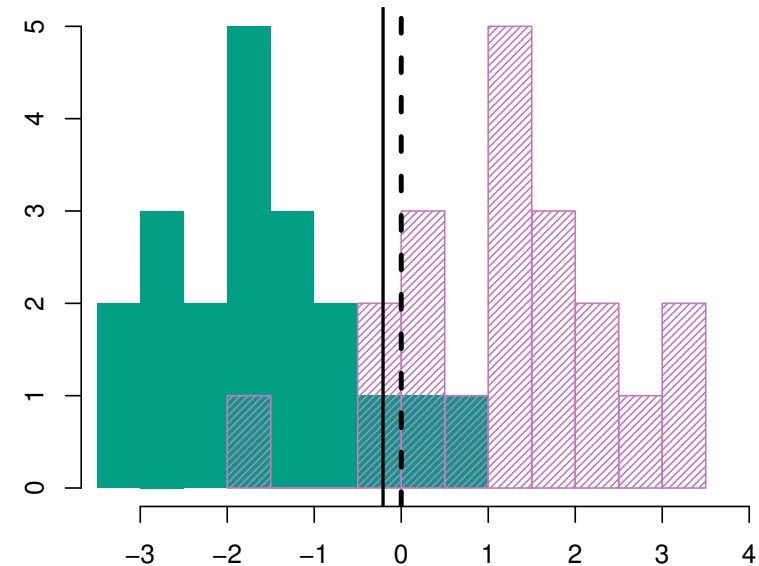
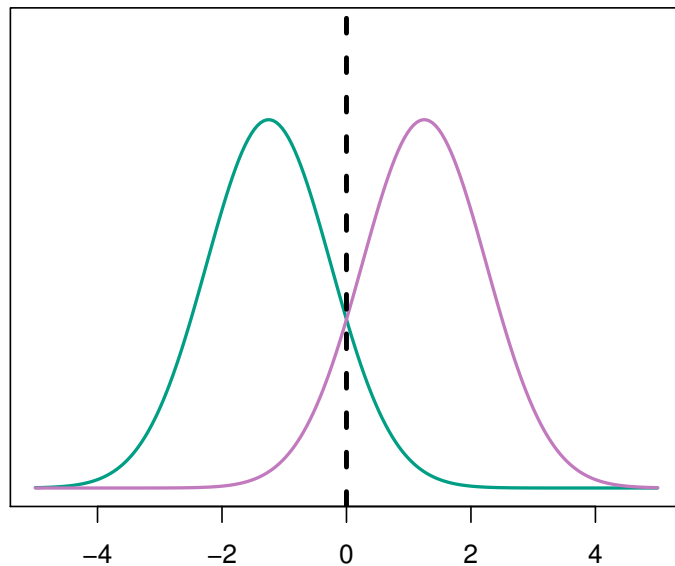
$$\delta_k(x) = x \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

- this is a linear function of $\mathbf{x}$,
- for $K = 2$ classes and $\pi_1 = \pi_2 = 0.5$, the decision boundary is at

$$x = \frac{\mu_1 + \mu_2}{2}$$

Estimating the parameters

- Typically these parameters are unknown, we estimate them from data,
- example below with $\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$ and $\sigma^2 = 1$



Linear discriminant analysis for $p > 1$

- Density: $f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}}e^{-\frac{1}{2}(\mathbf{x}-\mu)^T\Sigma^{-1}(\mathbf{x}-\mu)}$
- discriminant function: $\delta_k(\mathbf{x}) = \mathbf{x}^T\Sigma^{-1}\mu_k^T - \frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k + \log(\pi_k),$
- despite its complex form a linear function of $\mathbf{x}$.

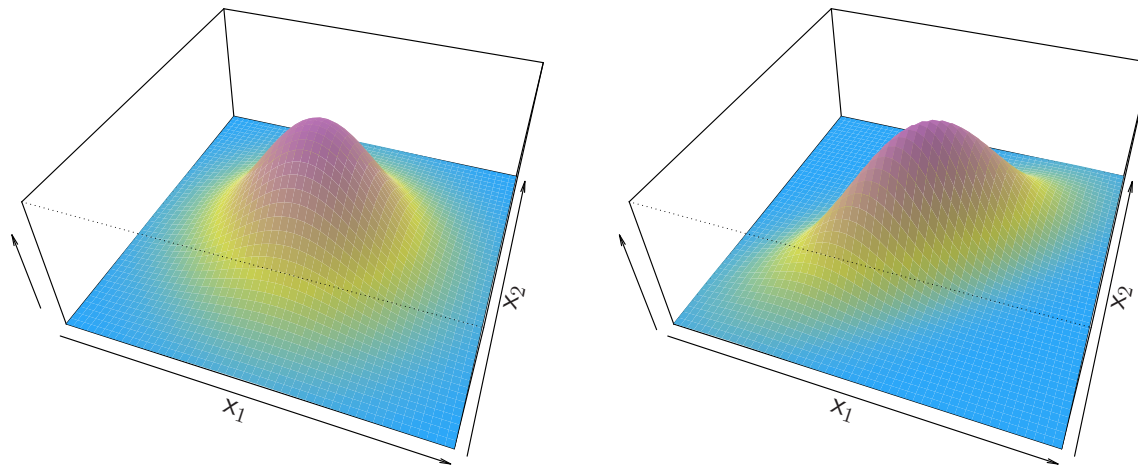
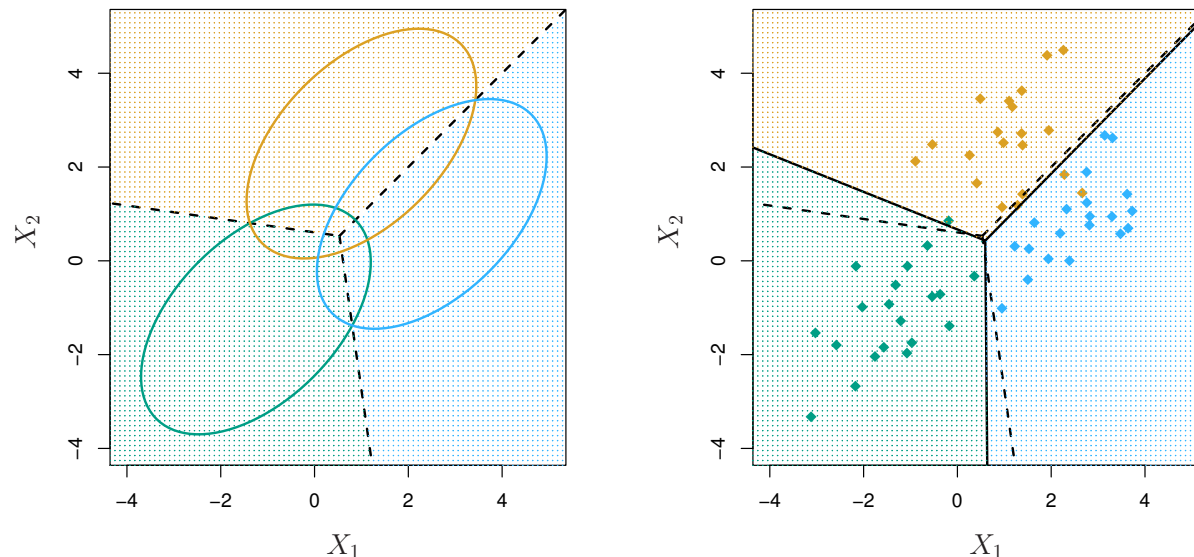


Illustration: $p = 2$ and $K = 3$ classes

- Three classes with the same priors, class-specific mean vectors and a **common covariance matrix**,
- ellipses in the left represent 95% confidence regions for each of the classes, dashed lines Bayes optimal decision boundaries,
- in the right LDA decision boundaries learned from a sample with 20 observations per class.



From $\delta_k(\mathbf{x})$ to class probabilities

- Turn discriminant scores into class probability estimates

$$\hat{Pr}(Y = k|X = x) = \frac{e^{\hat{\delta}_k(x)}}{\sum_{l=1}^K e^{\hat{\delta}_l(x)}}$$

- classifying to the largest $\delta_k(\mathbf{x})$ amounts to classifying to the class for which $Pr(Y = k|\mathbf{X} = \mathbf{x})$ is largest,
- when $K = 2$, classify to class 2 if $Pr(Y = 2|X = x) \geq 0.5$, else to class 1,
- **confusion matrix** and classification accuracy can be employed then

		<i>True Default Status</i>		
		No	Yes	Total
<i>Predicted Default Status</i>	No	9644	252	9896
	Yes	23	81	104
Total		9667	333	10000

Evaluation of a discriminative model

- This approach is often insufficient for
 - skewed classes (imbalanced class sizes),
 - unequal losses (different misclassification costs),
- for unequal losses, change the decision threshold from 0.5 to some other value from $[0,1]$
 - example: when predicting defaults in earlier Credit dataset, we would make around 75% error on the true Yes cases,
 - sensitivity is very low → changing the threshold adapts to a different loss function.

Unequal losses: Credit data

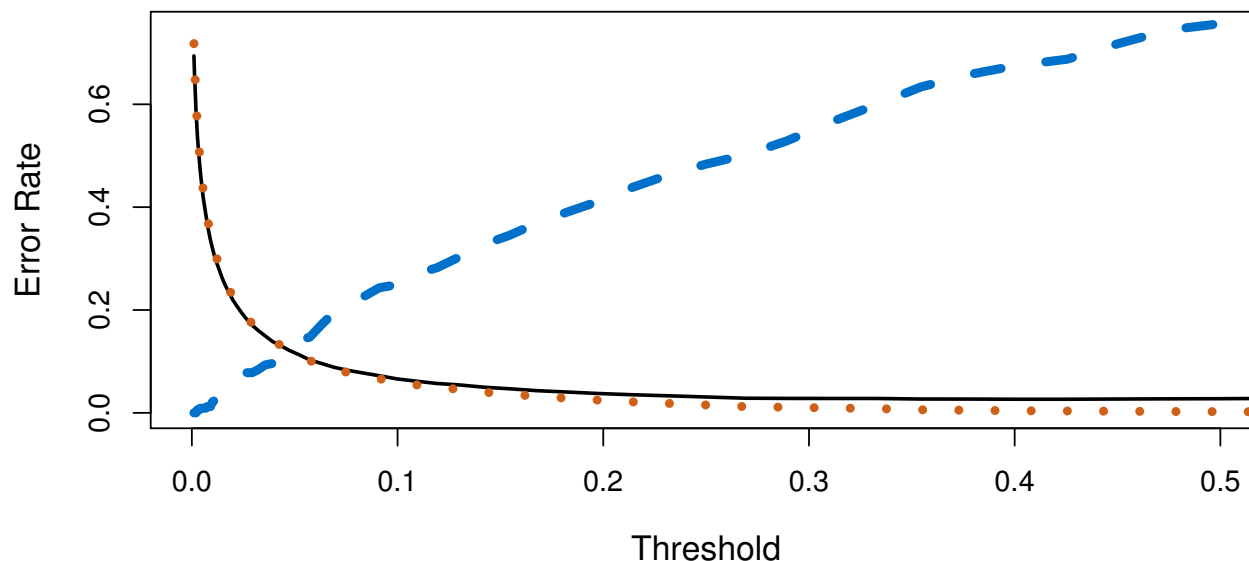
		True Default Status		
		No	Yes	Total
Predicted Default Status	No	9644	252	9896
	Yes	23	81	104
	Total	9667	333	10000

Annotations: TN (True Negative) points to 9644; FN (False Negative) points to 252; FP (False Positive) points to 23; TP (True Positive) points to 81. Below the table, 'N' is under 9667 and 'P' is under 333.

- Credit data with skewed classes, observations for LDA
 - $\frac{23+252}{10000}$ errors $\rightarrow$ a 2.75% misclassification rate!
 - overfitting not a big concern here since $n = 10000$ and $p = 4$,
 - if we always classified to class *No* in this case, we would make $\frac{333}{10000}$ errors, or only 3.33%,
 - of the true *No*'s, we make $\frac{23}{9667} = 0.2\%$ errors (false positive rate),
 - of the true *Yes*'s, we make $\frac{252}{333} = 75.7\%$ errors (false negative rate)!

Unequal losses: Credit data

- Let us change threshold in: if $Pr(Y = \text{default} | X = x) \geq \text{thres}$ then default
 - threshold 0.5 optimizes the overall error rate,
 - lower thresholds better fit the smaller class of defaulting customers (probably most interesting for a credit company).



Error rates as a function of the default threshold value, black solid ... the overall error, blue dashed ... the fraction of incorrectly classified defaulting customers (FNR), orange dotted ... the fraction of incorrectly classified non-defaulting customers (FPR).

Receiver operating characteristics (ROC)

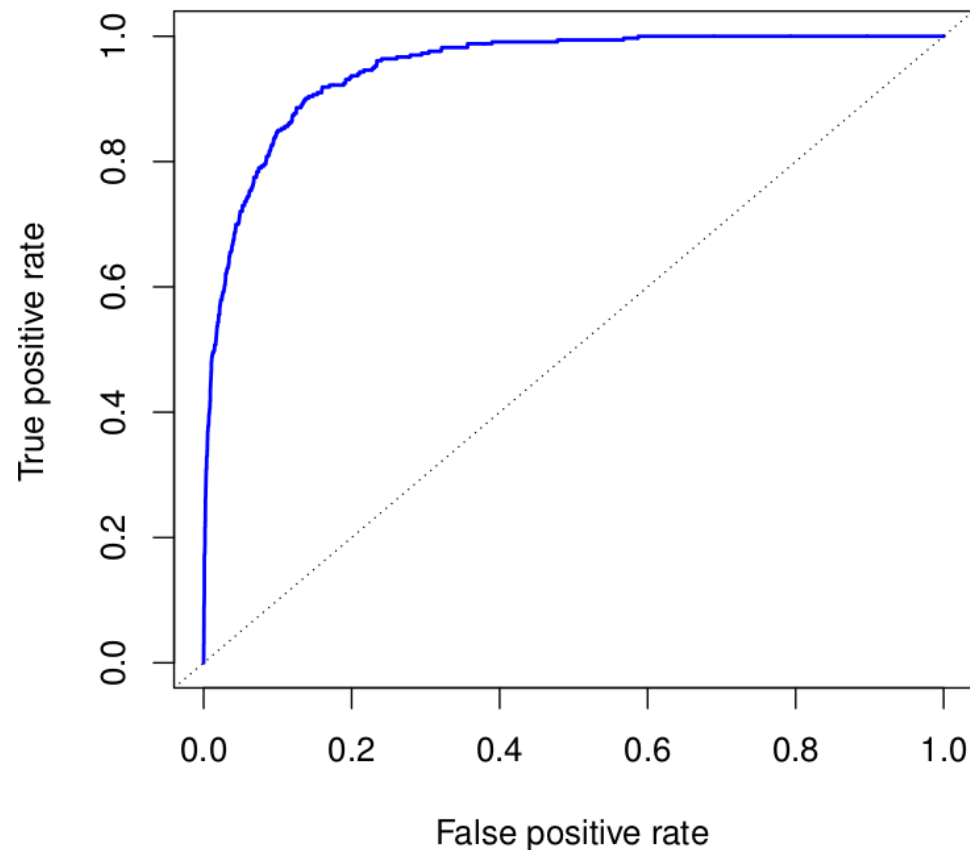
- Displays the errors for all possible thresholds
 - a popular way to evaluate probabilistic classifiers,
 - a better tool for imbalanced datasets than classification accuracy,
 - the overall performance of a classifier is given by the area under the ROC curve (AUC, AUROC), a number from $\langle 0, 1 \rangle$, 0.5 for random votes,
 - AUROC represents the probability that a random positive example is positioned to the right of a random negative example on the scale given by the probabilistic classifier.

$$TPR = \text{sensitivity} = \frac{\text{number of true positives}}{\text{total number of positives}} = \frac{TP}{P}$$

$$FPR = 1 - \text{specificity} = \frac{\text{number of false positives}}{\text{total number of negatives}} = \frac{FP}{N}$$

Receiver operating characteristics (ROC): Credit data

- AUC is 0.95, which is close to the maximum of 1,
- the LDA classifier can be considered very good.



A ROC curve traces out TPR and FPR as we vary the threshold value for the posterior probability of default.

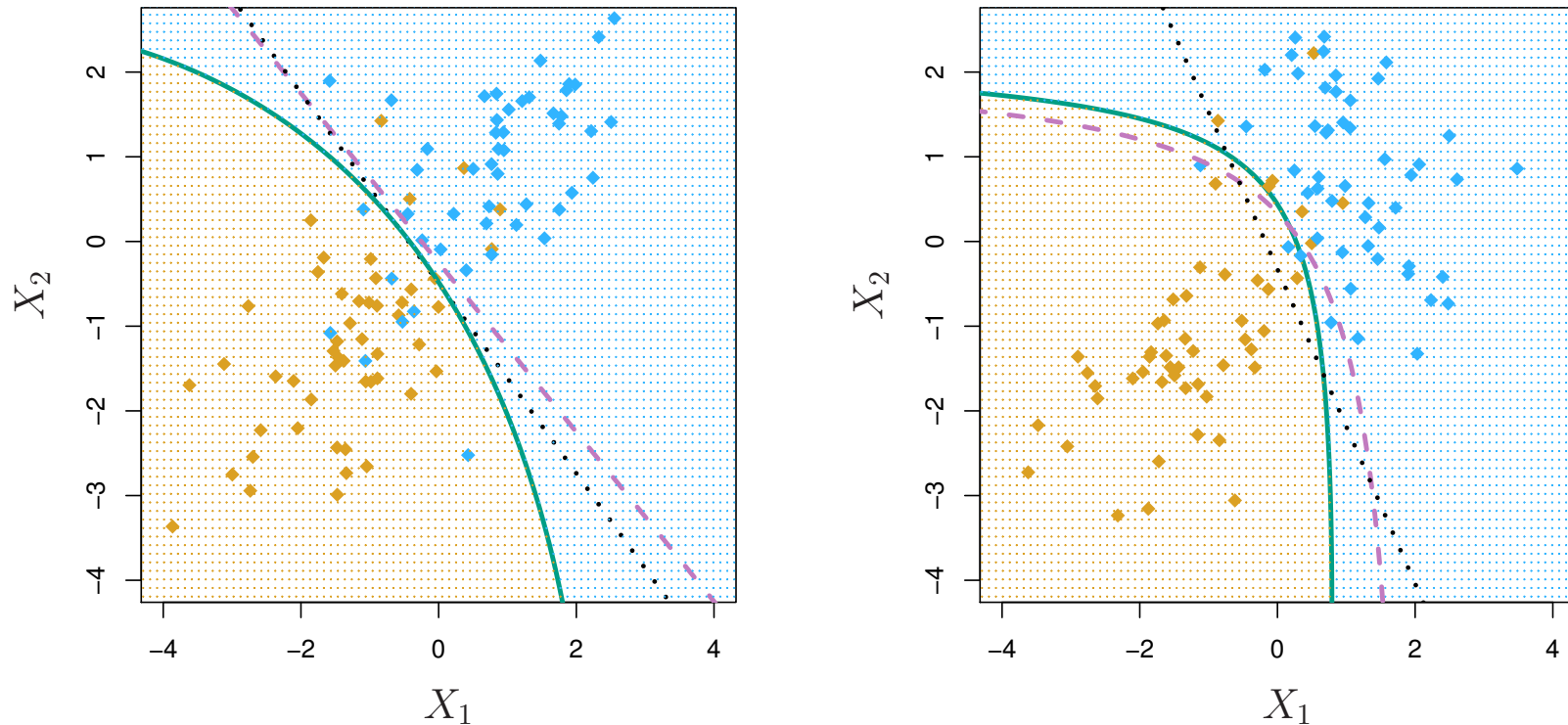
Other forms of discriminant analysis

- Consider different models in the general Bayes formula below

$$Pr(Y = k | \mathbf{X} = \mathbf{x}) = \frac{\pi_k f_k(\mathbf{x})}{\sum_{j=1}^K \pi_j f_j(\mathbf{x})}$$

- when $f_k(\mathbf{x})$ are Gaussian densities, with the same covariance matrix Σ in each class
 - linear discriminant analysis,
- with Gaussians but different Σ_k in each class
 - quadratic discriminant analysis,
- with $f_k(\mathbf{x}) = \prod_{j=1}^p f_{jk}(x_j)$ (conditional independence assumption) in each class
 - naïve Bayes classifier,
 - for Gaussian this means the Σ_k are diagonal.

Quadratic discriminant analysis



The Bayes optimal decision boundary in purple dashed line, LDA black dotted, QDA green solid. Left: the covariance matrices truly match, LDA is close to optimal solution, QDA suffers from higher variance. Right: the orange class has a positive correlation between predictors, the blue class negative, class covariance matrices differ, the optimal boundar is quadratic, LDA suffers from higher bias.

Summary

- Logistic regression is very popular for classification, especially when $K = 2$,
- LDA is useful when
 - the number of samples is small, or the classes are well separated, and Gaussian assumptions are reasonable,
 - $K > 2$, because it also provides low-dimensional views of the data,
- naïve Bayes is useful when
 - the dimension is very large,
- kNN is useful when
 - the parametric assumptions used above do not hold,
 - the decision boundary is highly non-linear,
 - disadvantage: no direct outcome on feature importance,
- none of the methods dominates the others in every situation.

The main references

:: Resources (slides, scripts, tasks) and reading

- G. James, D. Witten, T. Hastie and R. Tibshirani: **An Introduction to Statistical Learning with Applications in R**. Springer, 2014.
- K. Markham: **In-depth Introduction to Machine Learning in 15 hours of Expert Videos**. Available at R-bloggers.